

Experiment -1

Linux Programming Environment & Basic Commands

A) Setting up of Linux Programming Environment

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

***Online:**

Cocalc.ai –

1. Open Cocalc.ai - Go to the online cocalc.ai website and create or login your account .
2. Create a new project – Start a new Cocalc project where you can store or run your C program.
3. Open a terminal – Open the terminal inside your project to compile or execute C program.
4. Create a C file – Create a file with .c extension such as program.c , and write your code in it.
5. Write your code – Enter your code using standard C syntax, such as #include<stdio.h> .
6. Compile your program - In the Terminal, use 'gcc program.c -o program' to convert your C source code into an executable program.
7. Run the program – Type './program' in the terminal to execute the compiled C program to see the output .
8. Save your work – Save your .c file in Cocalc project you can access or modify your project later.

***Offline:**

Methods: Dual boot, Single boot, Bootable Pendrive, VM

- (i) Dual Boot – It means to have two operating systems on one computer. In our case one is windows and the other is Linux.
- (ii) Single Boot – Single boot means having only one operating system on the computer.
- (iii) Bootable Pendrive – It is a USB FLASH DRIVE that has all the necessary files needed to start or install the operating system on the computer.
- (iv) VM – It is a software based computer that runs inside your operating system.

1. Virtual Machine Manager (Hypervisor)

Type-1 (Baremetal)

It is a physical computer server or hardware system where software or an operating system runs directly on the components.

E.g. 1. Game Servers

2. A.I. Training

3. Financial Trading

Type-2 (Hosted)

It is a virtualization software that runs on top of the existing operating system. E.g.

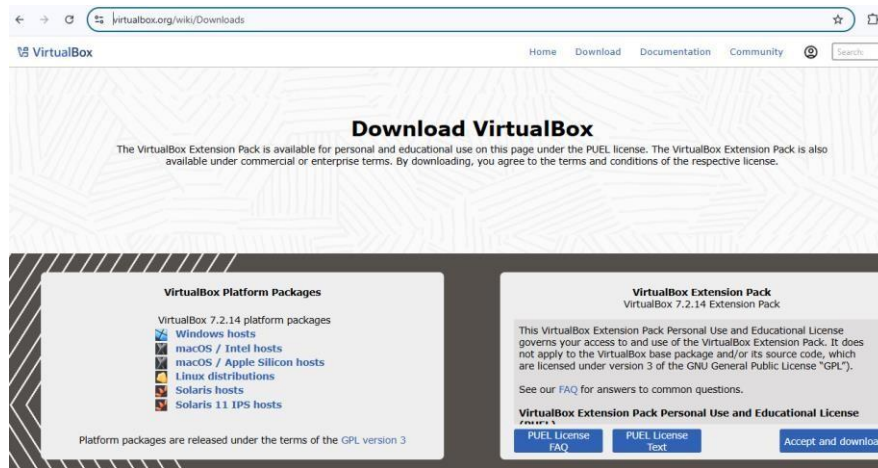
1. Oracle Virtualbox

2. VM Ware

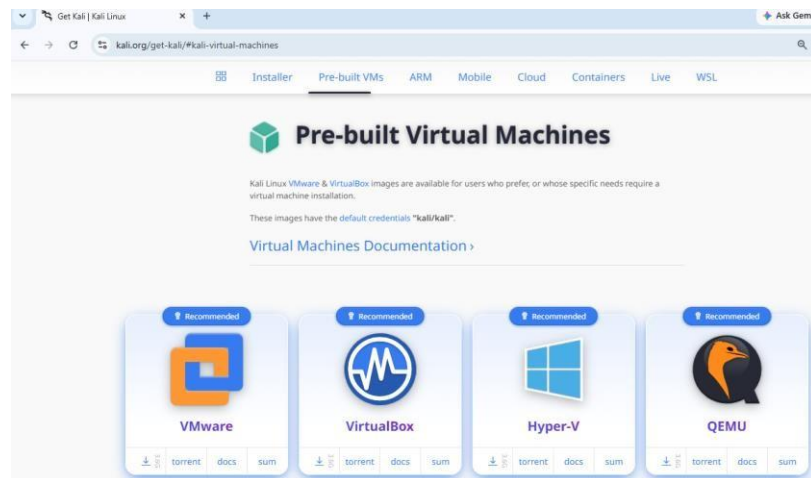
We would be using Oracle Virtualbox.

2. Installation of VMM & VM

Step-1: Download Oracle Virtual Box for “Windows hosts”.

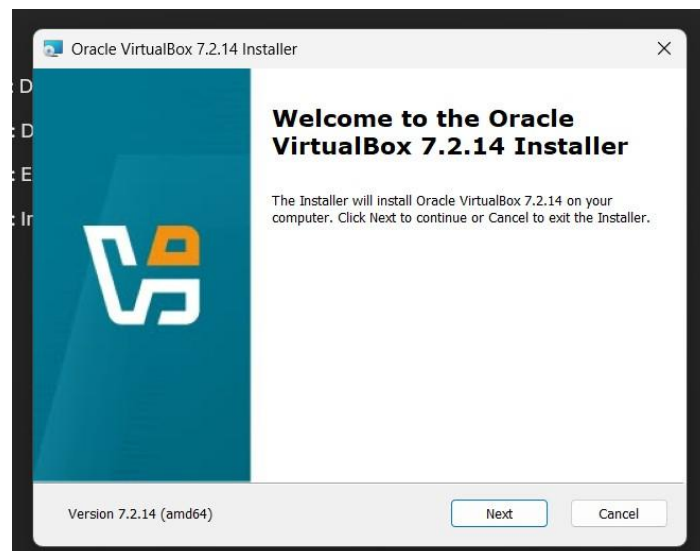


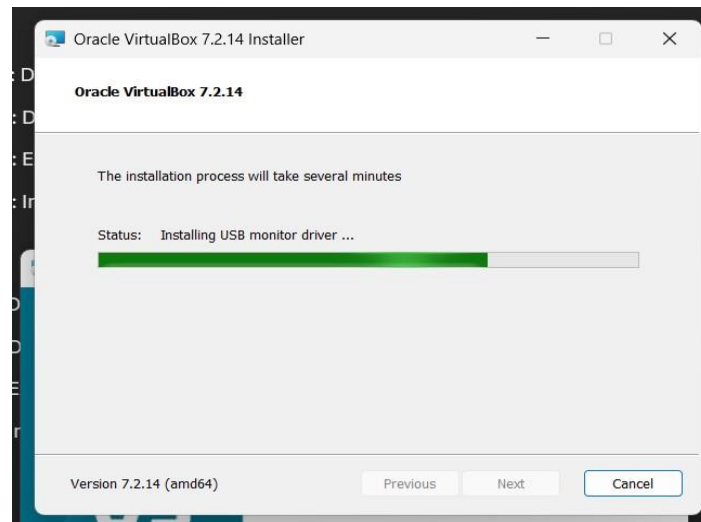
Step-2: Download Pre-built Kali Linux VM for Virtualbox.



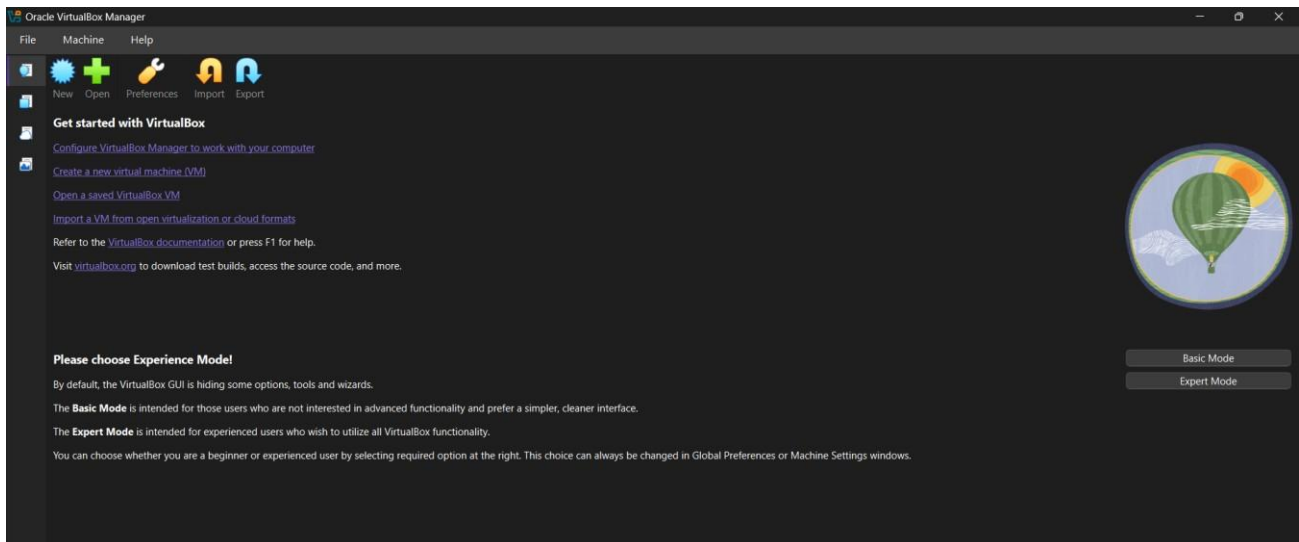
Step-3: Extracted the Kali Linux downloaded file

Step-4: Install Virtualbox

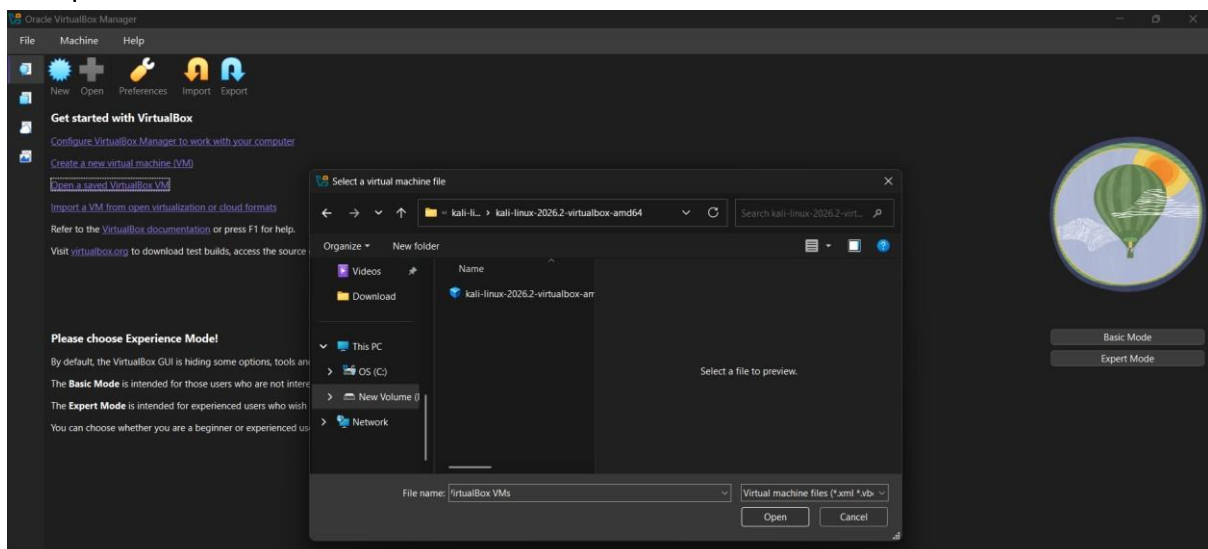


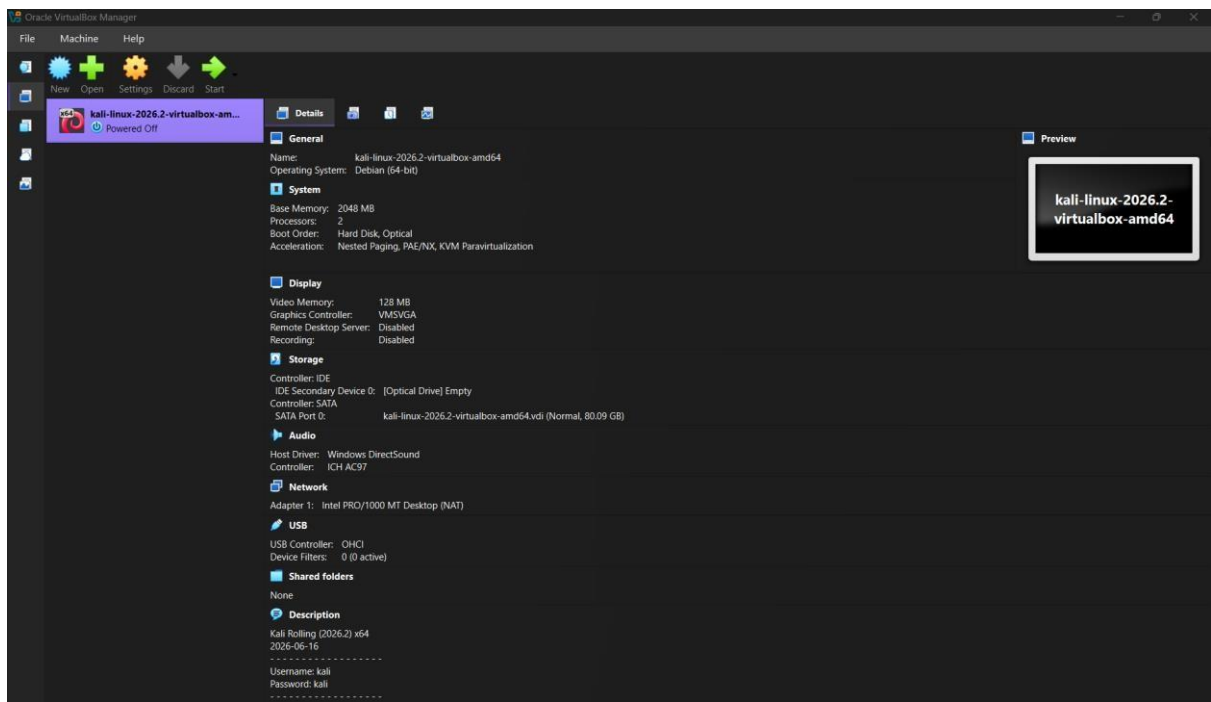


Step-5: Open Virtual box

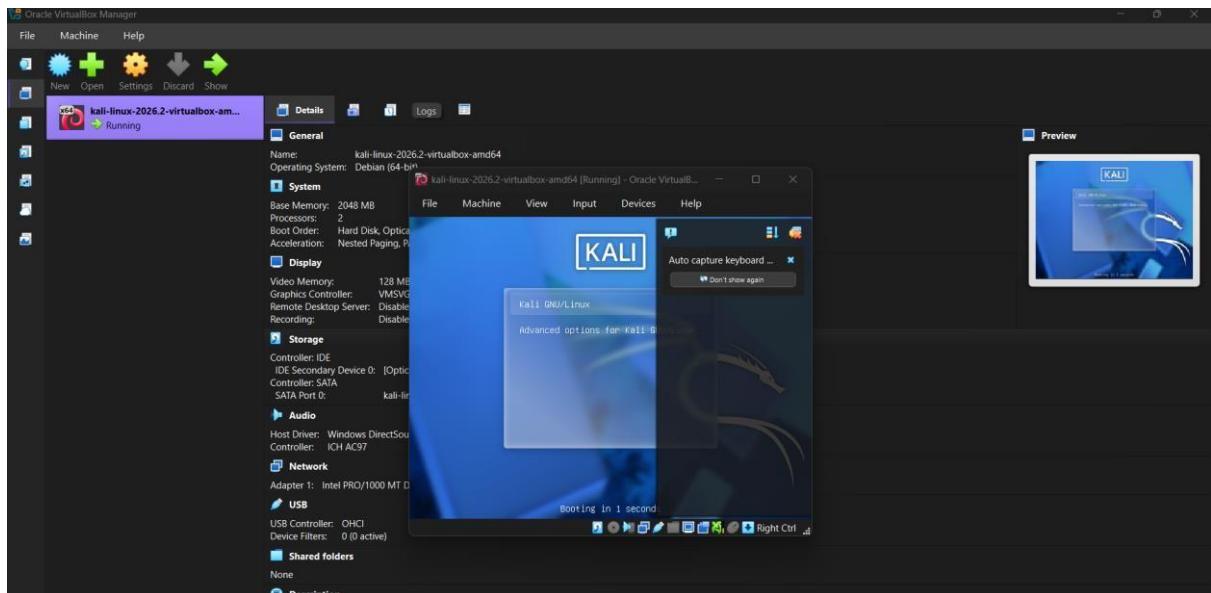


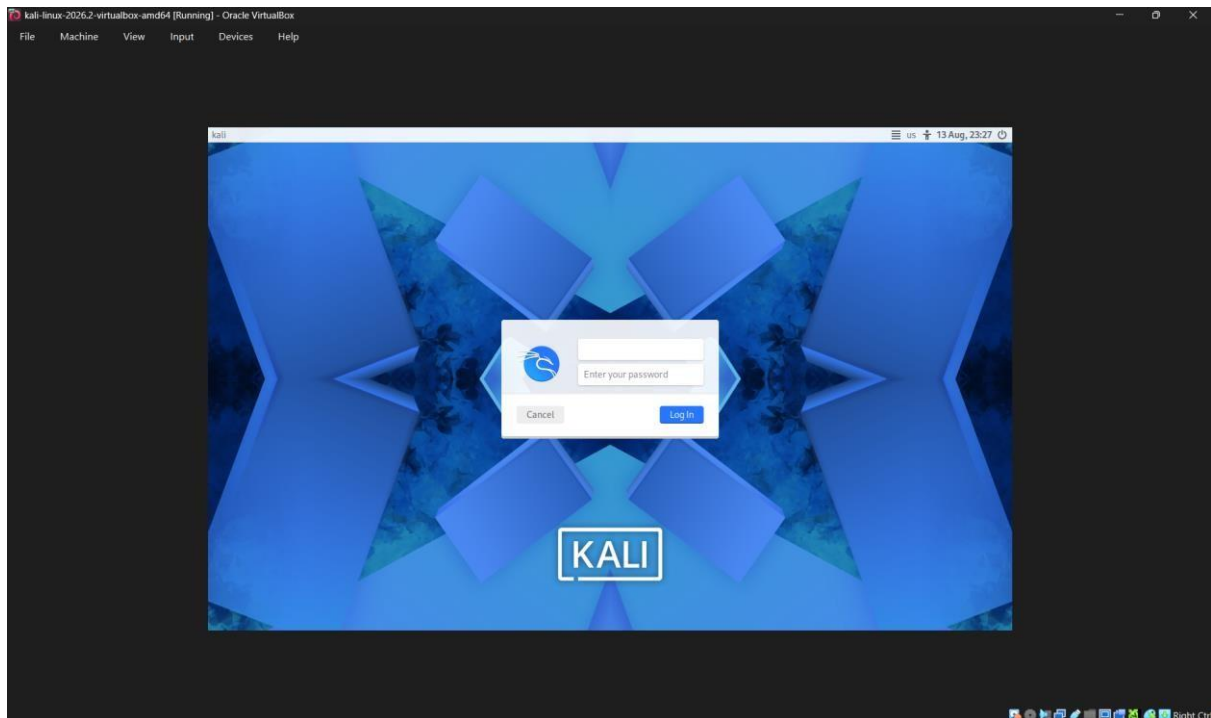
Step-6: Open the “saved VM”





Step-7: Start the VM





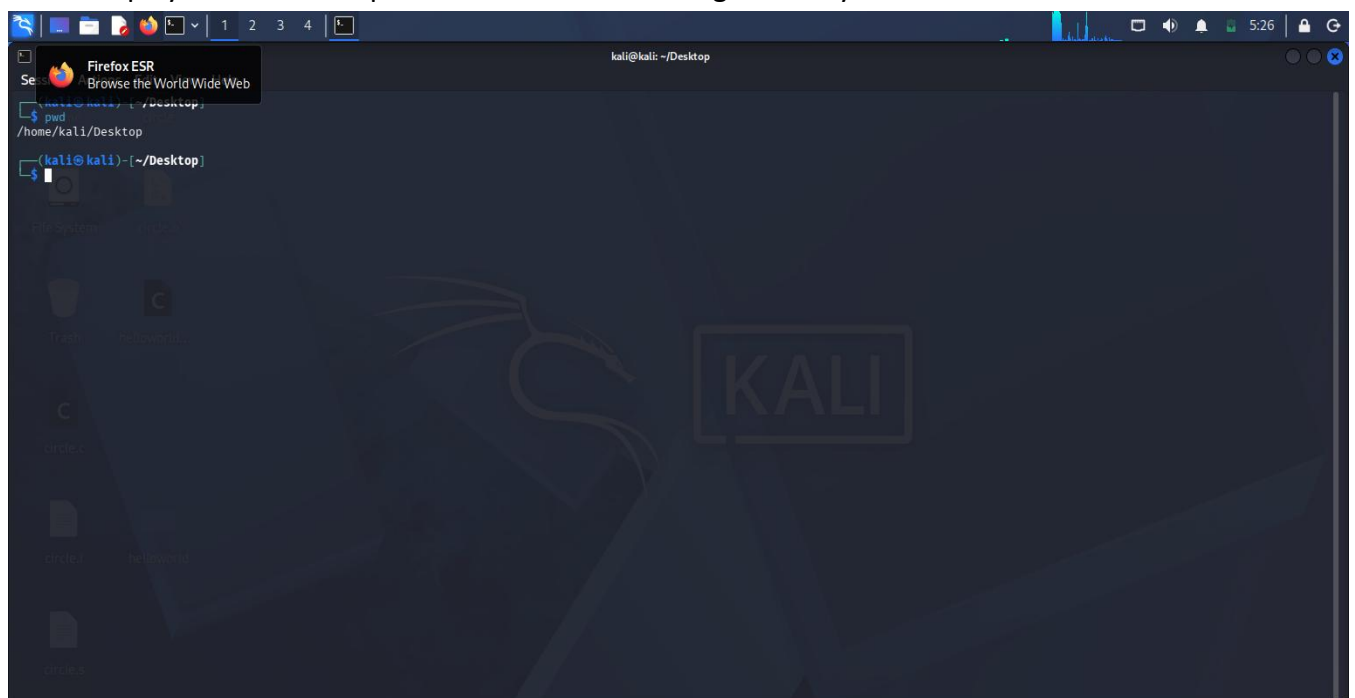
Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.

Step-8: Open terminal and create the following simple program, compile and execute to understand the working of the environment.

B) Working with Linux Commands

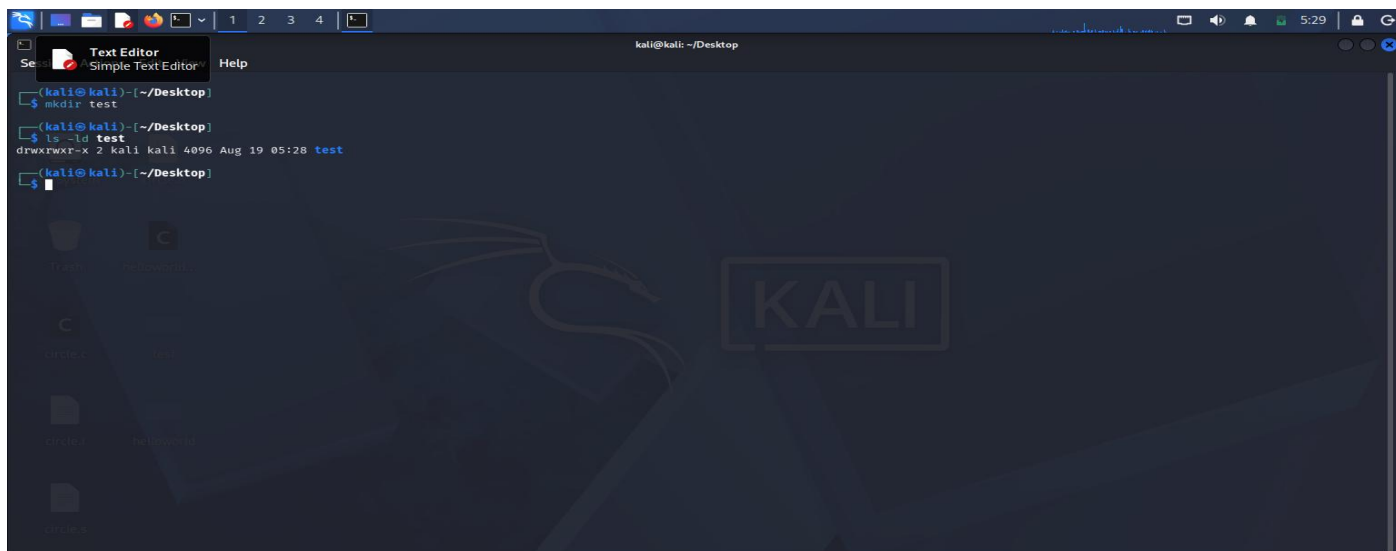
1. Pwd :-

Brief: Displays the absolute path of the current working directory.



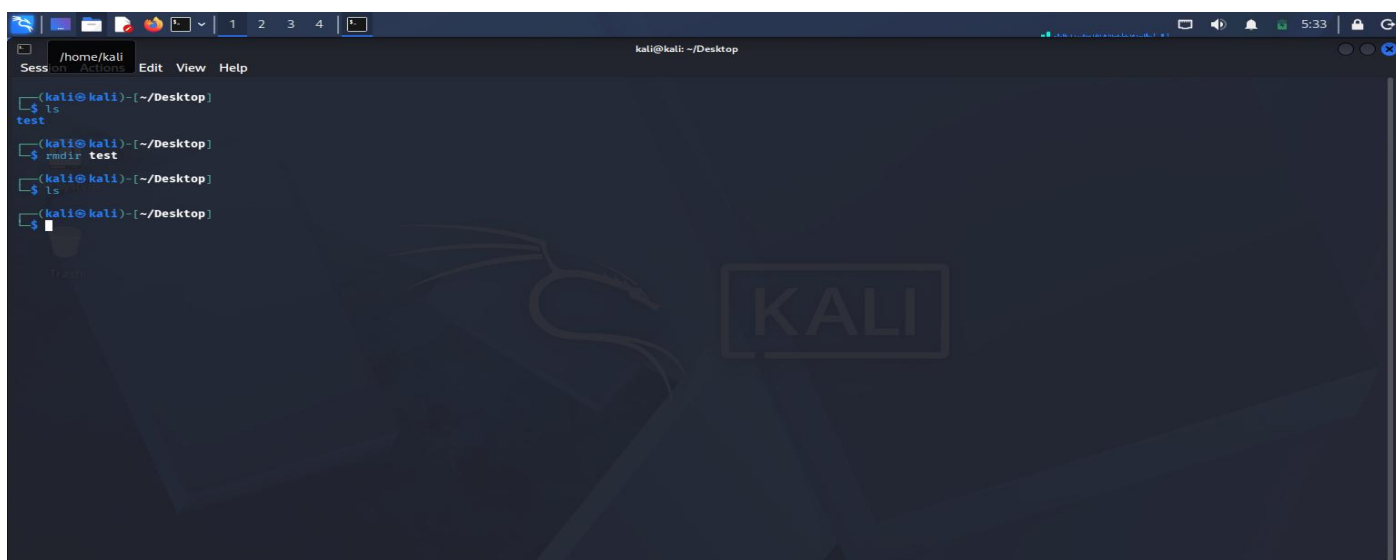
2. Mkdir :-

Brief: Creates a new directory.



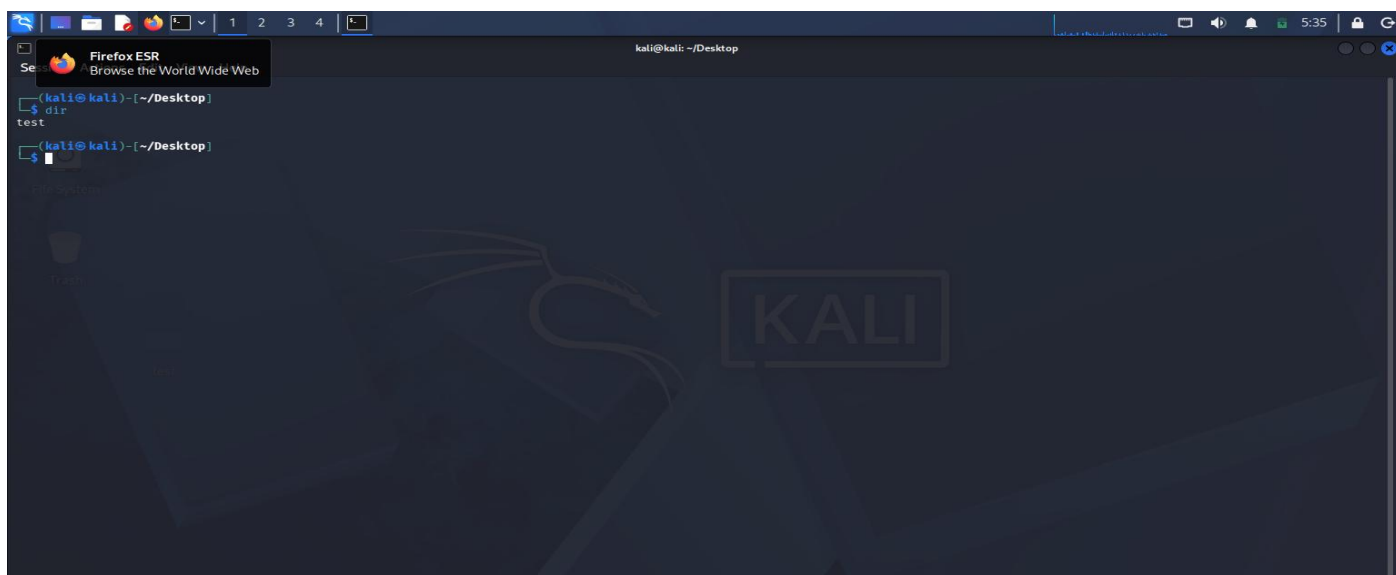
3. Rmdir :-

Brief: Removes an empty directory.



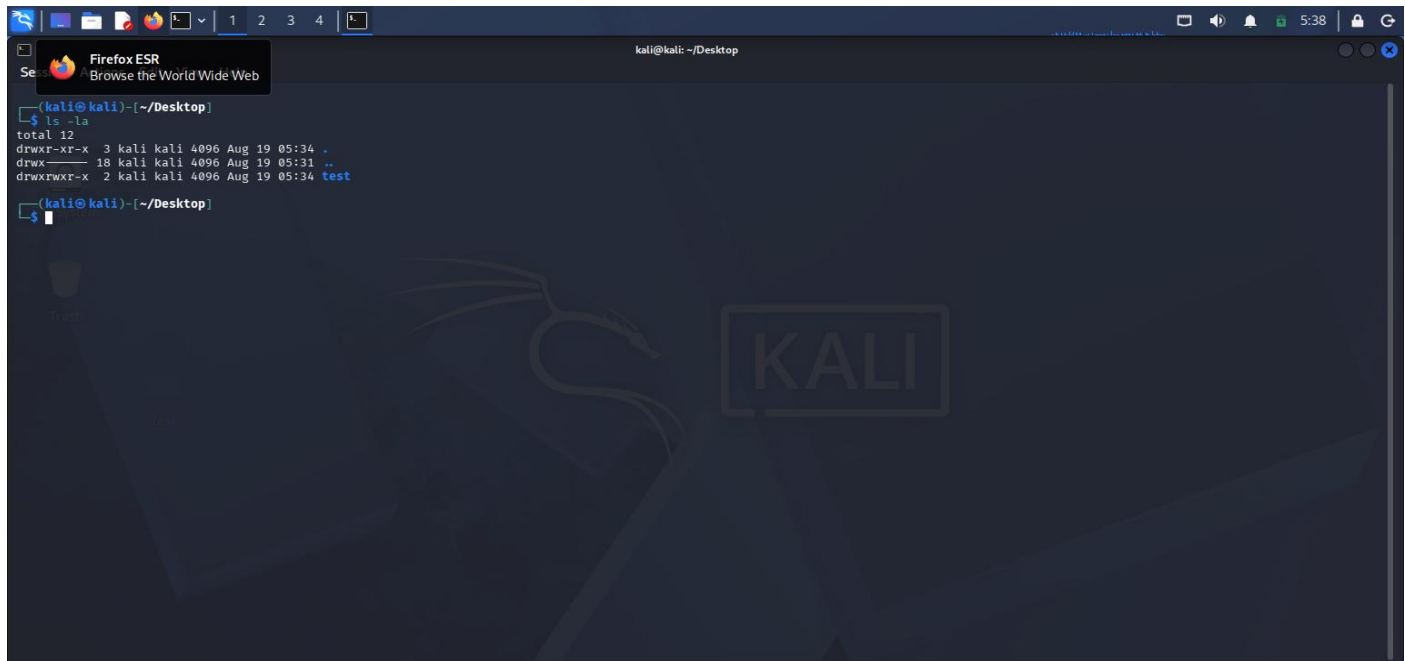
4. dir :-

Brief: Lists files and directories in the current location.



5. ls :-

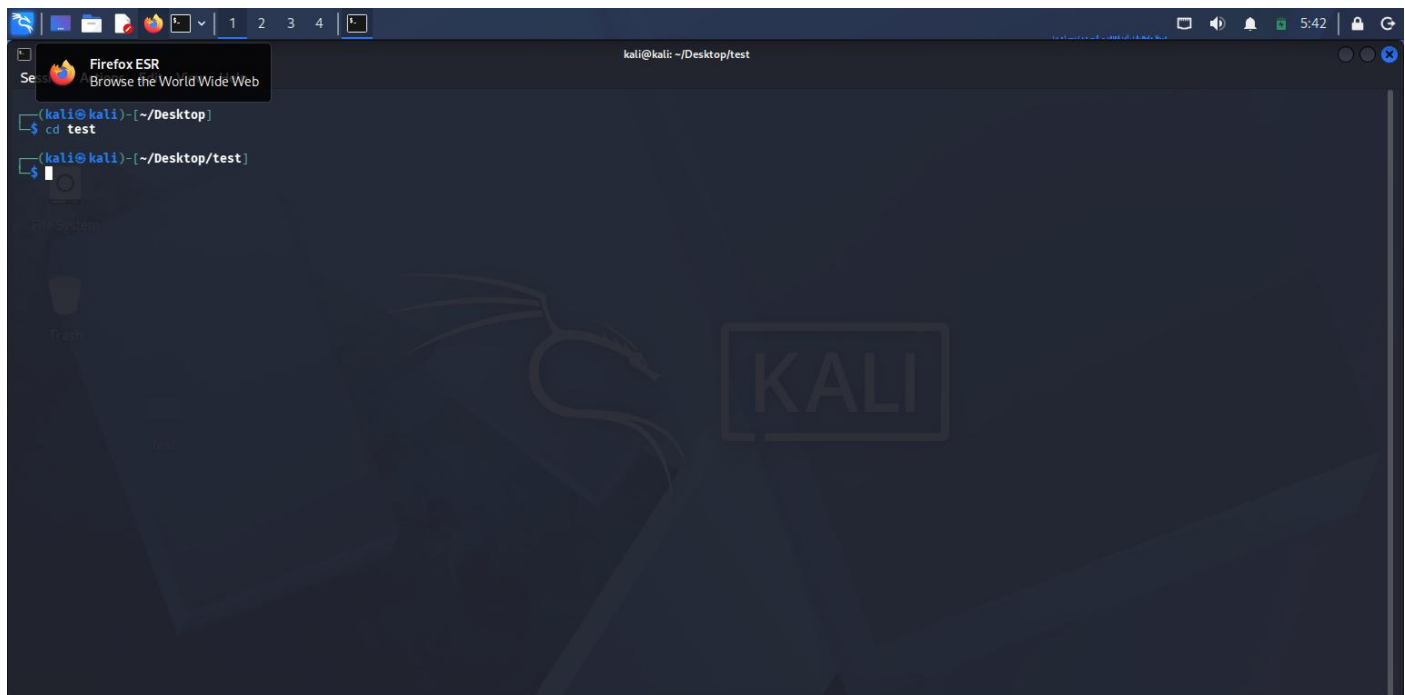
Brief: Lists files and directories. It can also display detailed information.



```
(kali@kali)-[~/Desktop]
$ ls -la
total 12
drwxr-xr-x  3 kali kali 4096 Aug 19 05:34 .
drwx----- 18 kali kali 4096 Aug 19 05:31 ..
drwxrwxr-x  2 kali kali 4096 Aug 19 05:34 test
```

6. cd :-

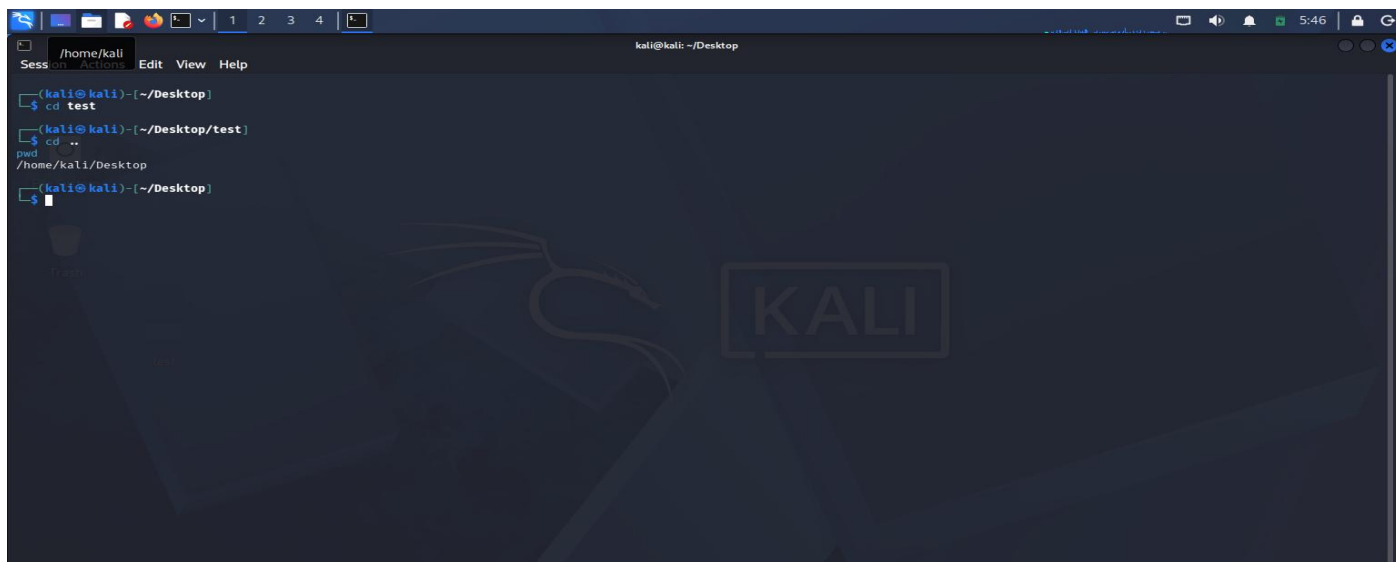
Brief: Changes the current working directory.



```
(kali@kali)-[~/Desktop]
$ cd test
(kali@kali)-[~/Desktop/test]
$
```

7. cd .. :-

Brief: Moves one level up to the parent directory.

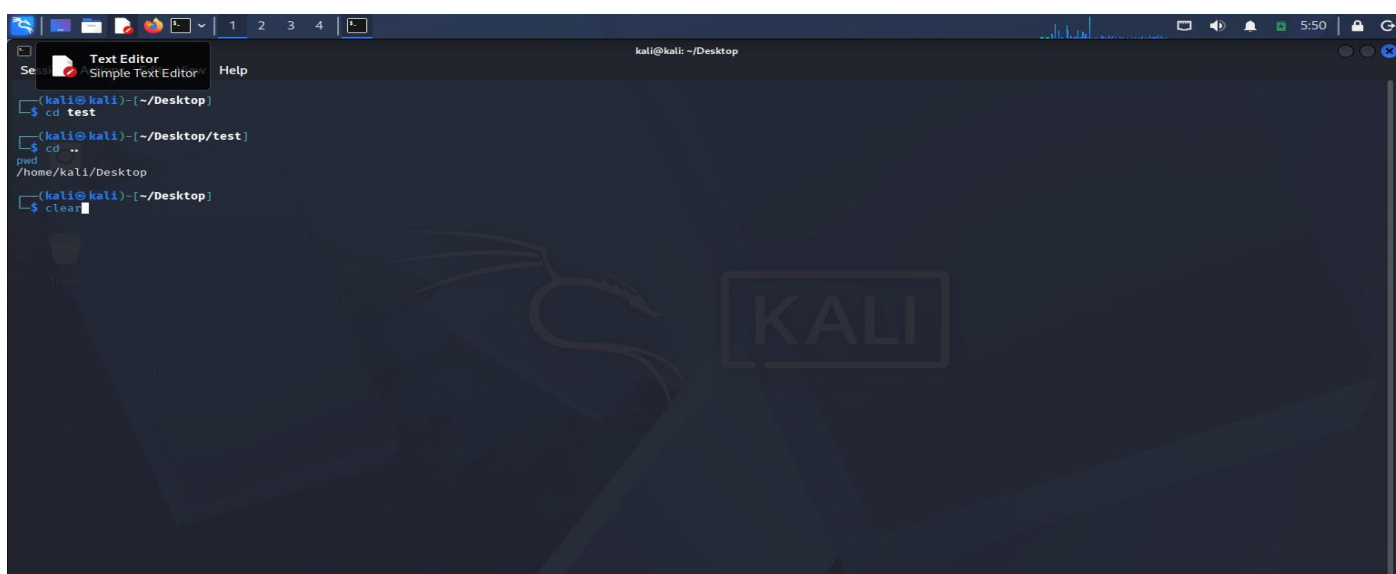


```
kali@kali: ~/Desktop
$ cd test
$ cd ..
$ pwd
/home/kali/Desktop
$
```

The terminal window shows a series of commands to navigate the file system. It starts in the ~/Desktop directory, moves into a subdirectory named 'test', then back to the parent directory with 'cd ..'. The 'pwd' command confirms the current directory is /home/kali/Desktop. The prompt is ready for the next command.

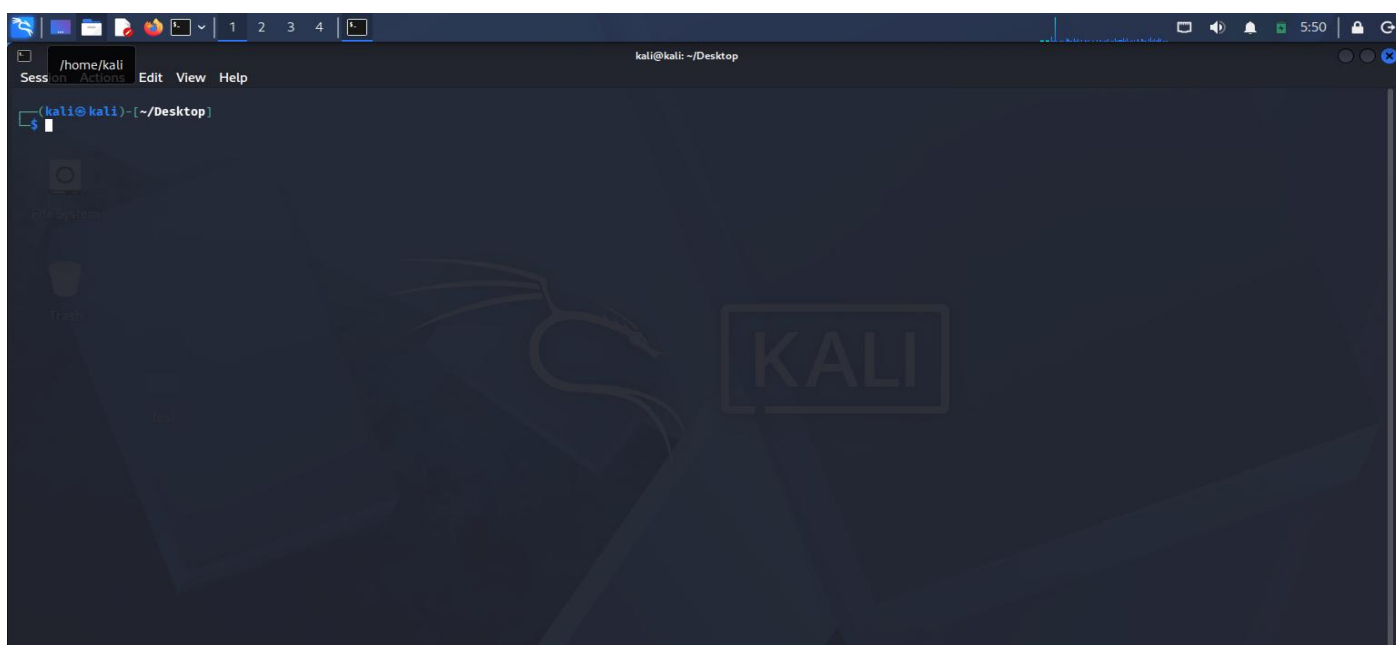
8. clear :-

Brief: Clears the visible contents of the terminal.



```
kali@kali: ~/Desktop
$ cd test
$ cd ..
$ pwd
/home/kali/Desktop
$ clear
```

The terminal window shows the 'clear' command being entered at the prompt. The previous directory navigation commands and their outputs are still visible in the terminal history.

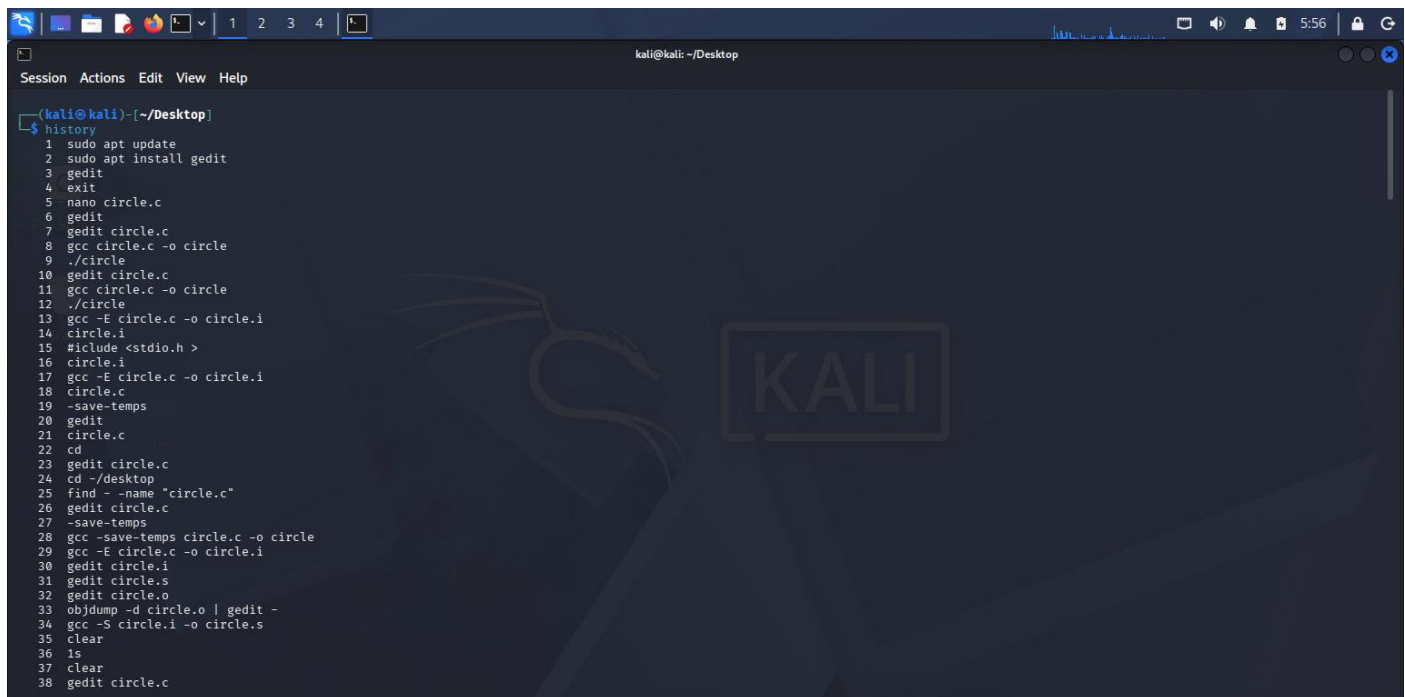


```
kali@kali: ~/Desktop
$
```

The terminal window shows the result of the 'clear' command. The screen has been cleared, and only the current prompt 'kali@kali: ~/Desktop \$' is visible. The terminal history is no longer shown.

9. history :-

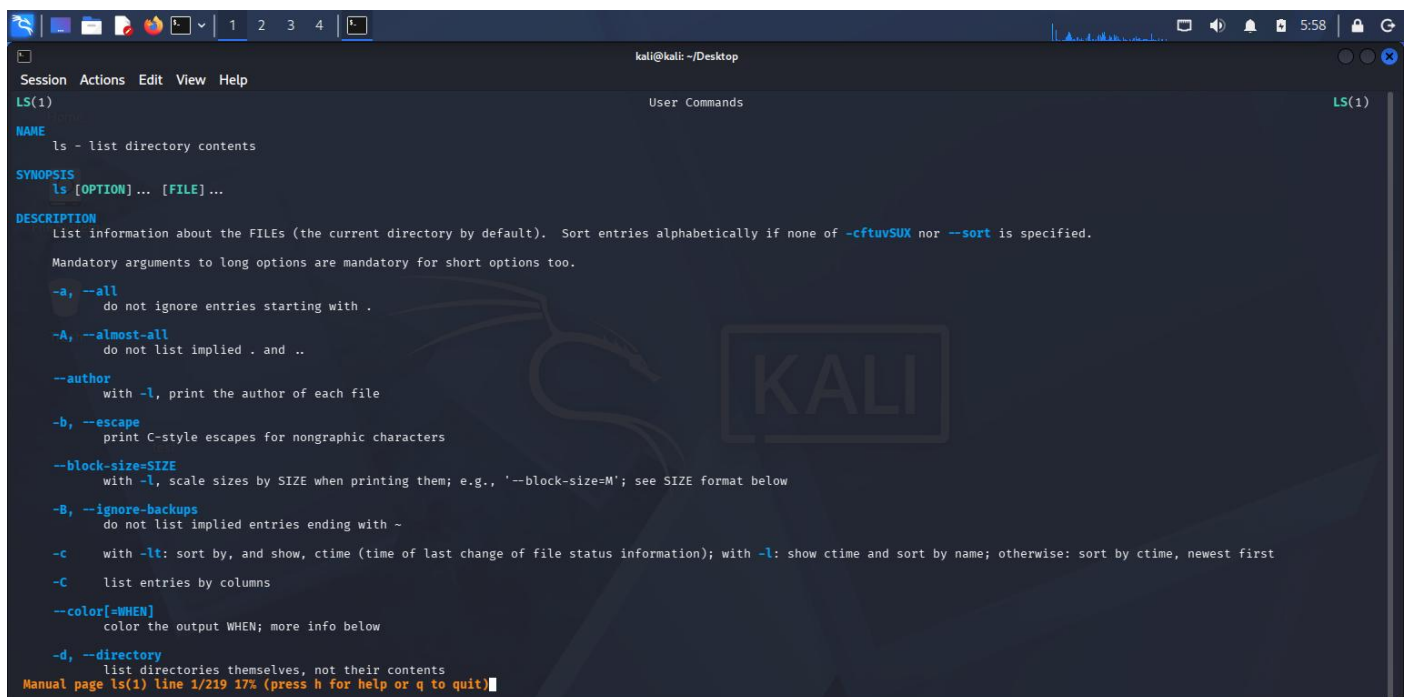
Brief: Displays previously executed commands.



```
(kali@kali)-[~/Desktop]
└─$ history
1 sudo apt update
2 sudo apt install gedit
3 gedit
4 exit
5 nano circle.c
6 gedit
7 gedit circle.c
8 gcc circle.c -o circle
9 ./circle
10 gedit circle.c
11 gcc circle.c -o circle
12 ./circle
13 gcc -E circle.c -o circle.i
14 circle.i
15 #include <stdio.h >
16 circle.i
17 gcc -E circle.c -o circle.i
18 circle.c
19 -save-temps
20 gedit
21 circle.c
22 cd
23 gedit circle.c
24 cd ~/desktop
25 find - -name "circle.c"
26 gedit circle.c
27 -save-temps
28 gcc -save-temps circle.c -o circle
29 gcc -E circle.c -o circle.i
30 gedit circle.i
31 gedit circle.s
32 gedit circle.o
33 obfdump -d circle.o | gedit -
34 gcc -S circle.i -o circle.s
35 clear
36 ls
37 clear
38 gedit circle.c
```

10. man :-

Brief: Displays the manual page and usage information for a command.



```
LS(1) User Commands LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [OPTION]... [FILE]...

DESCRIPTION
  List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

  Mandatory arguments to long options are mandatory for short options too.

  -a, --all
      do not ignore entries starting with .

  -A, --almost-all
      do not list implied . and ..

  --author
      with -l, print the author of each file

  -b, --escape
      print C-style escapes for nongraphic characters

  --block-size=SIZE
      with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups
      do not list implied entries ending with ~

  -c
      with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first

  -C
      list entries by columns

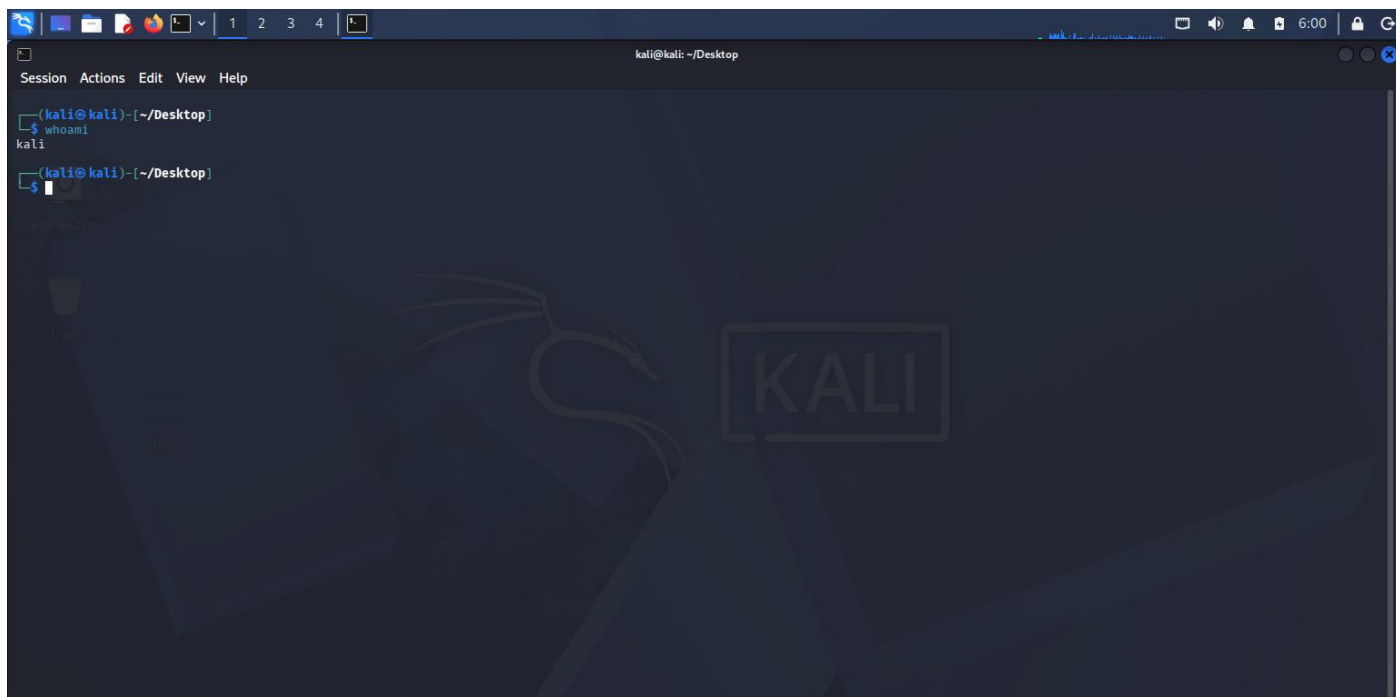
  --color[=WHEN]
      color the output WHEN; more info below

  -d, --directory
      list directories themselves, not their contents

Manual page ls(1) line 1/219 17% (press h for help or q to quit)
```

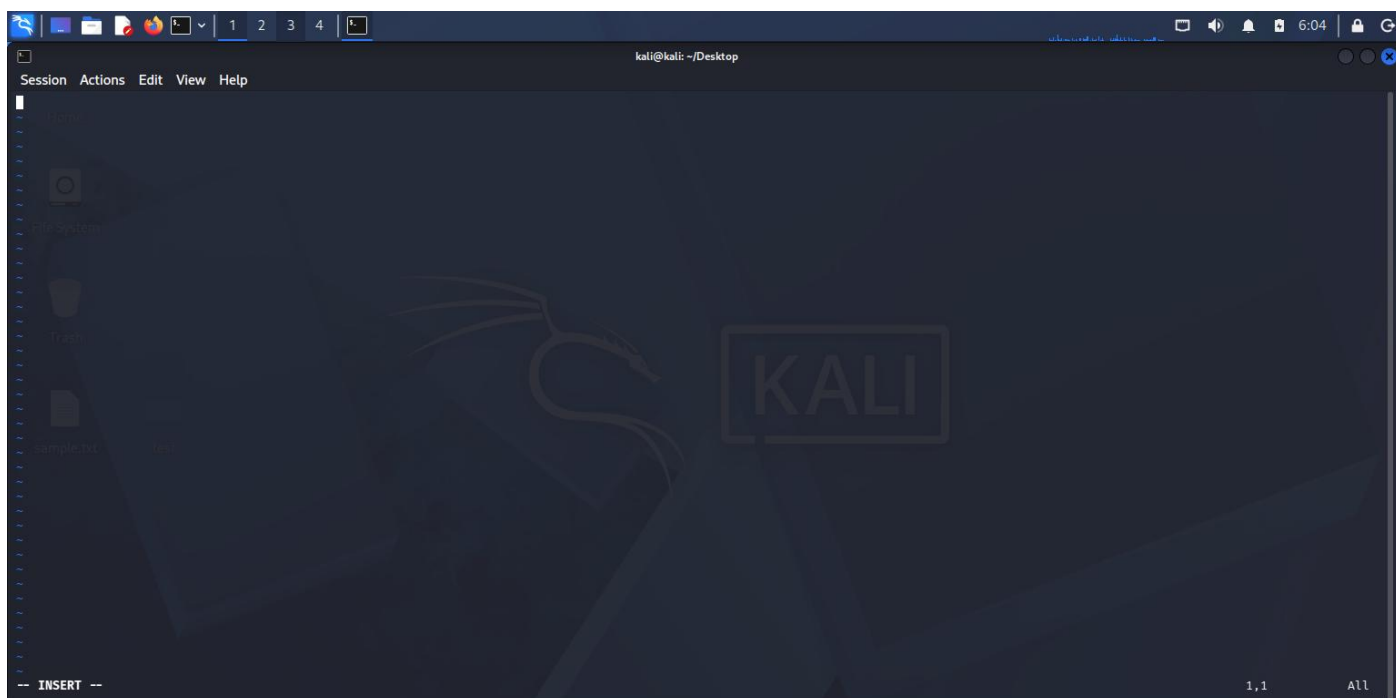
11. whoami :-

Brief: Displays the username of the currently logged-in user.



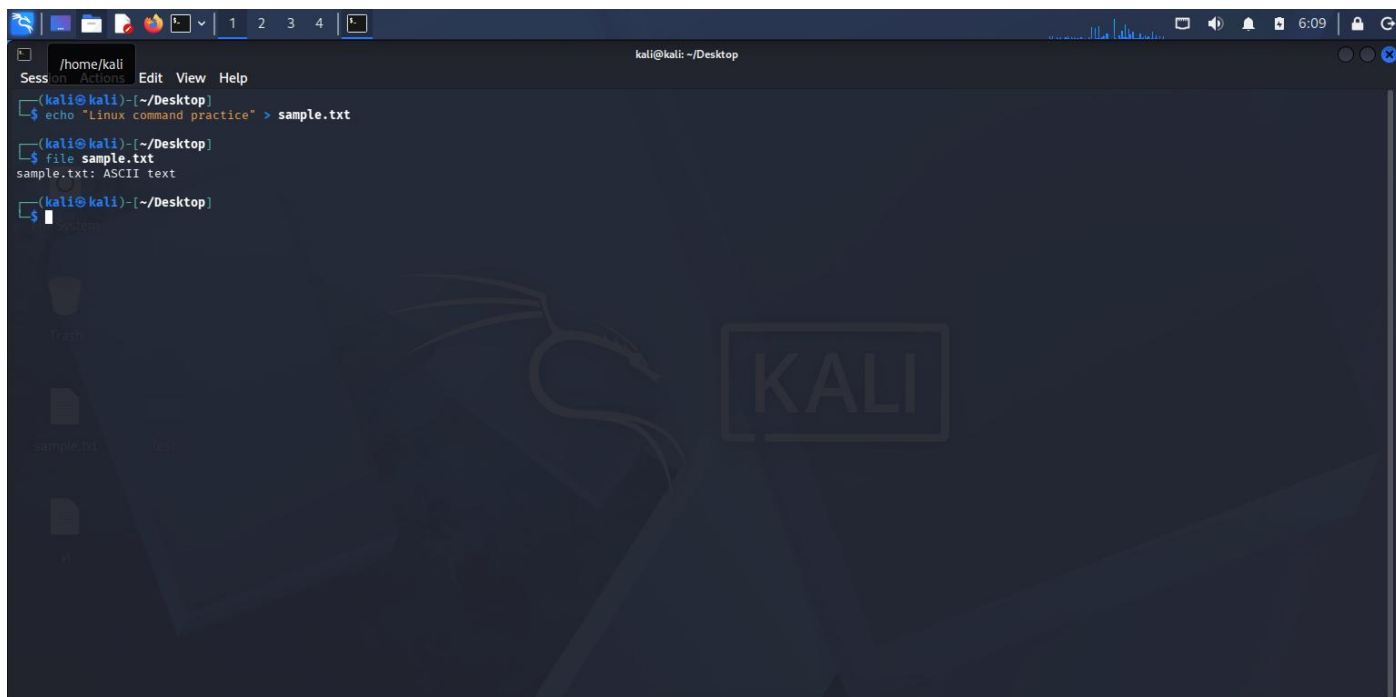
12. vi :-

Brief: A terminal-based text editor used to create and modify files.



13. file :-

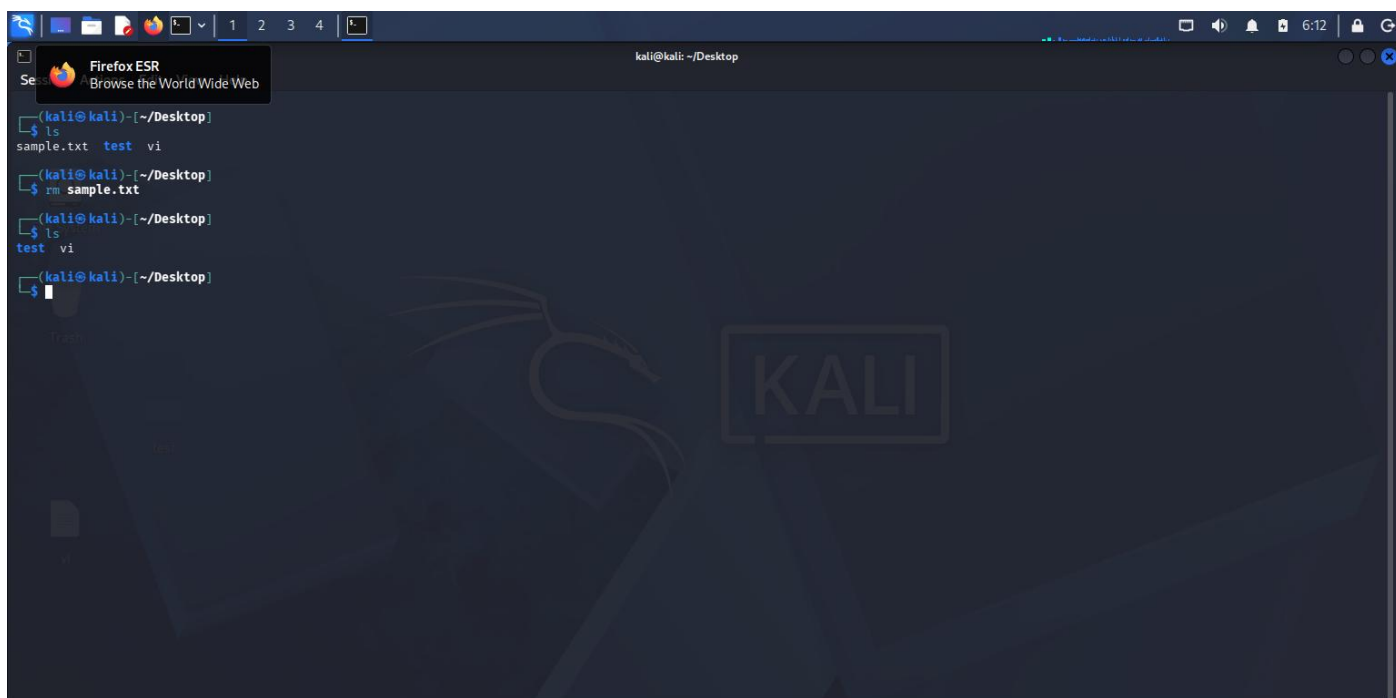
Brief: Identifies the type and format of a file.

A terminal window on a Kali Linux desktop. The terminal shows the user creating a file named 'sample.txt' using the 'echo' command, then verifying its existence and content with the 'file' command. The desktop background features a Kali Linux logo and the word 'KALI' in a box.

```
kali@kali: ~/Desktop
$ echo "Linux command practice" > sample.txt
$ file sample.txt
sample.txt: ASCII text
```

14. rm :-

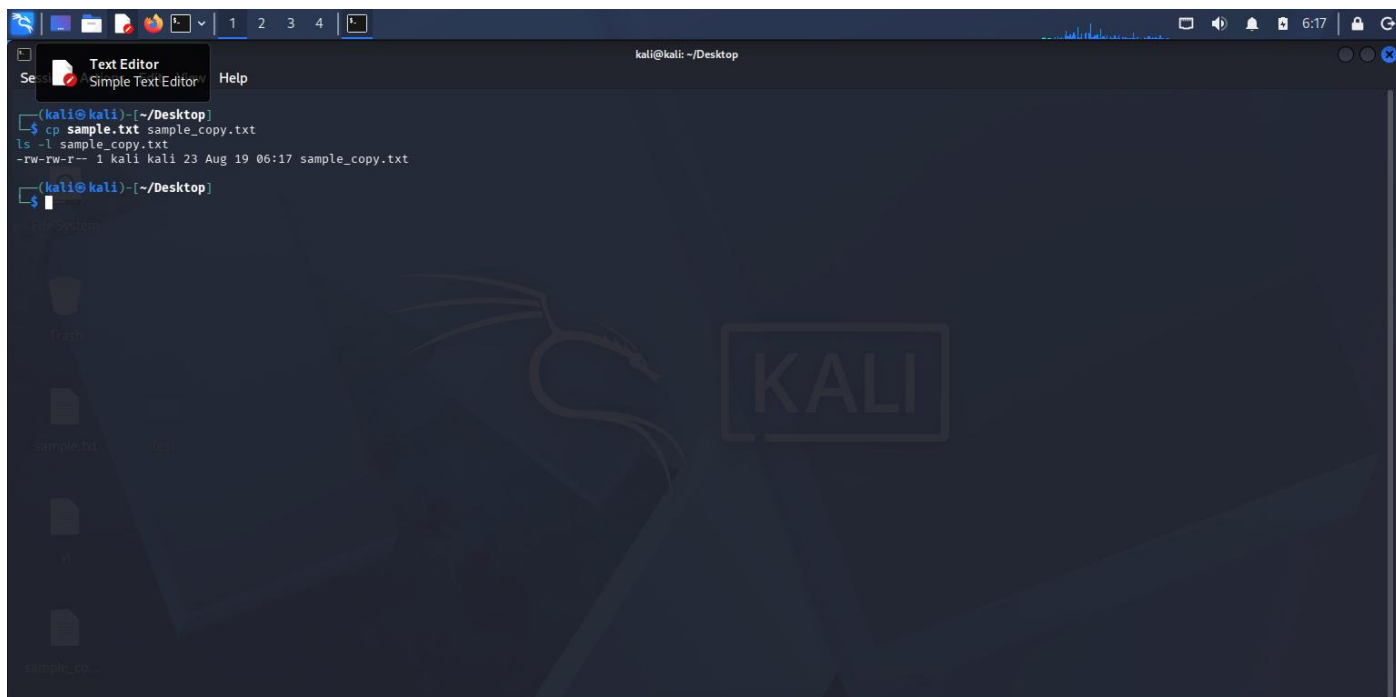
Brief: Deletes a file or directory when appropriate options are used.

A terminal window on a Kali Linux desktop. The terminal shows the user listing files, creating a file named 'test', and then deleting it using the 'rm' command. The desktop background features a Kali Linux logo and the word 'KALI' in a box.

```
kali@kali: ~/Desktop
$ ls
sample.txt  test  vi
$ rm sample.txt
$ ls
test  vi
```

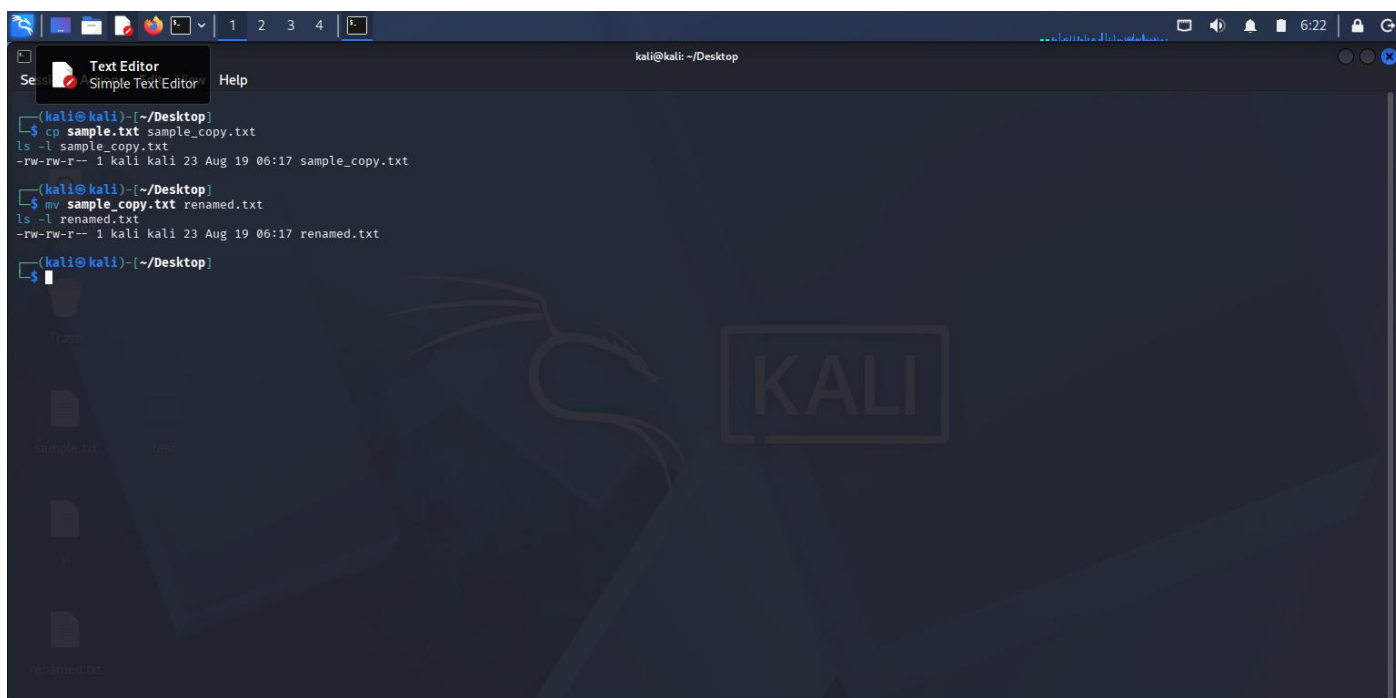
15. cp :-

Brief: Copies a file from one location to another.



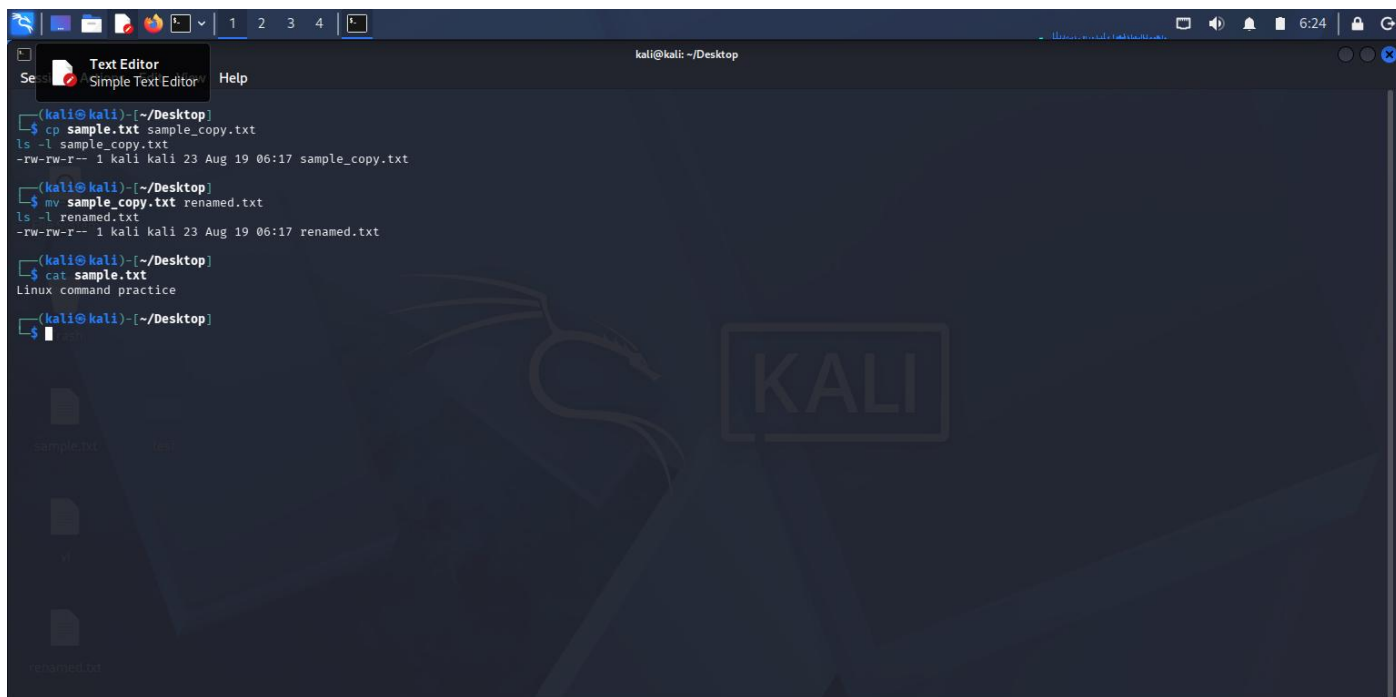
16. mv :-

Brief: Moves a file or renames it.



17. cat :-

Brief: Displays the complete contents of a file.



A screenshot of a Kali Linux terminal window. The window title is "kali@kali: ~/Desktop". The terminal shows the following commands and output:

```
(kali@kali)-[~/Desktop]
$ cp sample.txt sample_copy.txt
ls -l sample_copy.txt
-rw-rw-r-- 1 kali kali 23 Aug 19 06:17 sample_copy.txt

(kali@kali)-[~/Desktop]
$ mv sample_copy.txt renamed.txt
ls -l renamed.txt
-rw-rw-r-- 1 kali kali 23 Aug 19 06:17 renamed.txt

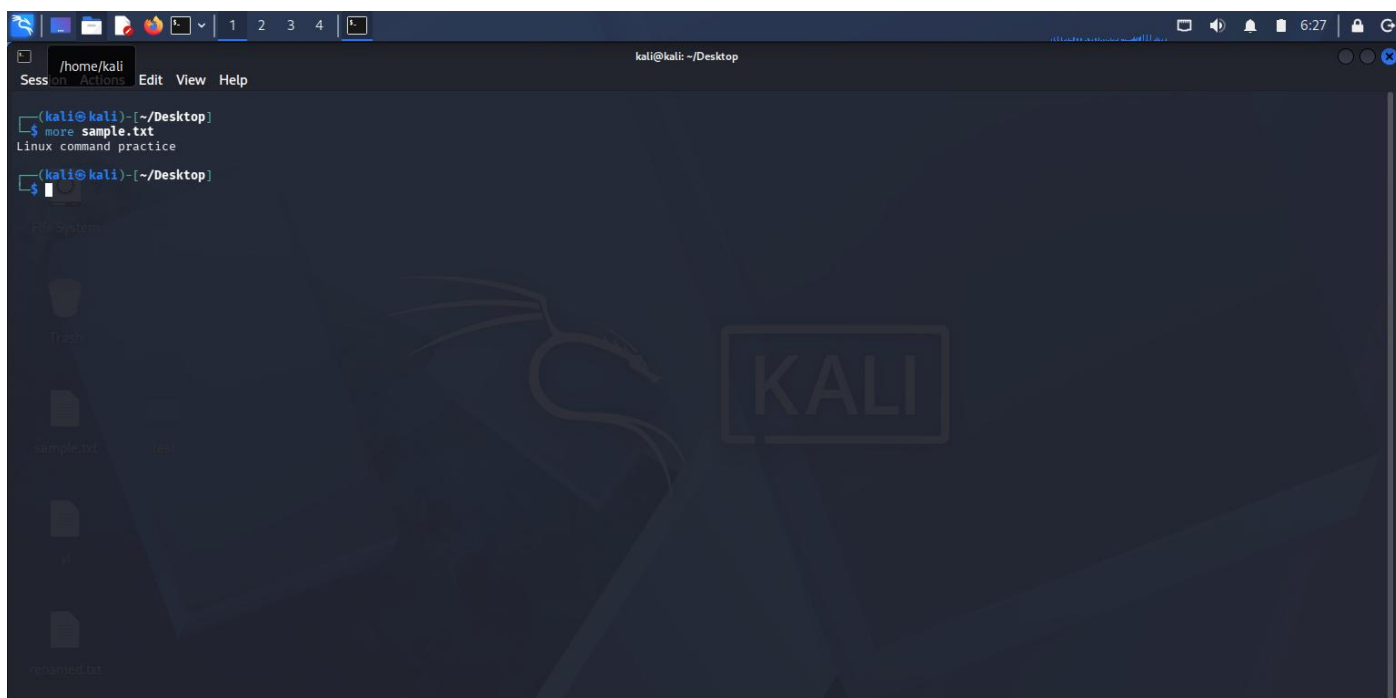
(kali@kali)-[~/Desktop]
$ cat sample.txt
Linux command practice

(kali@kali)-[~/Desktop]
$
```

The desktop background features a Kali Linux logo and the word "KALI". Files "sample.txt", "sample_copy.txt", and "renamed.txt" are visible on the desktop.

18. more :-

Brief: Displays a file one screen at a time.



A screenshot of a Kali Linux terminal window. The window title is "kali@kali: ~/Desktop". The terminal shows the following commands and output:

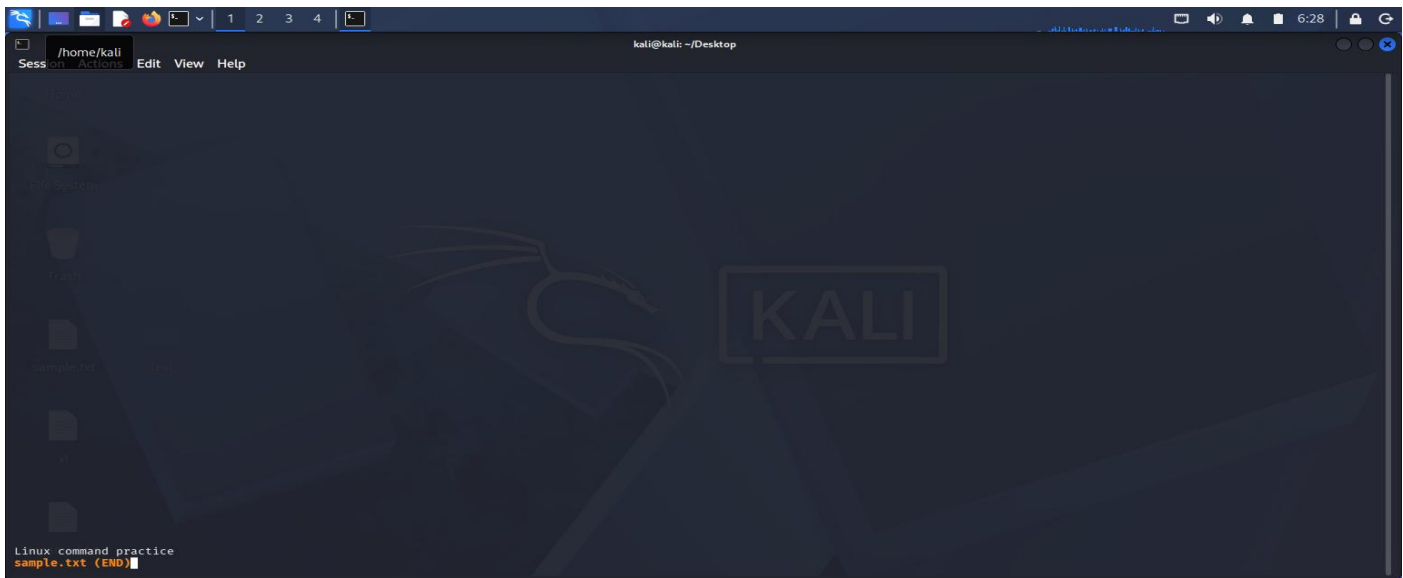
```
(kali@kali)-[~/Desktop]
$ more sample.txt
Linux command practice

(kali@kali)-[~/Desktop]
$
```

The desktop background features a Kali Linux logo and the word "KALI". Files "sample.txt", "sample_copy.txt", and "renamed.txt" are visible on the desktop.

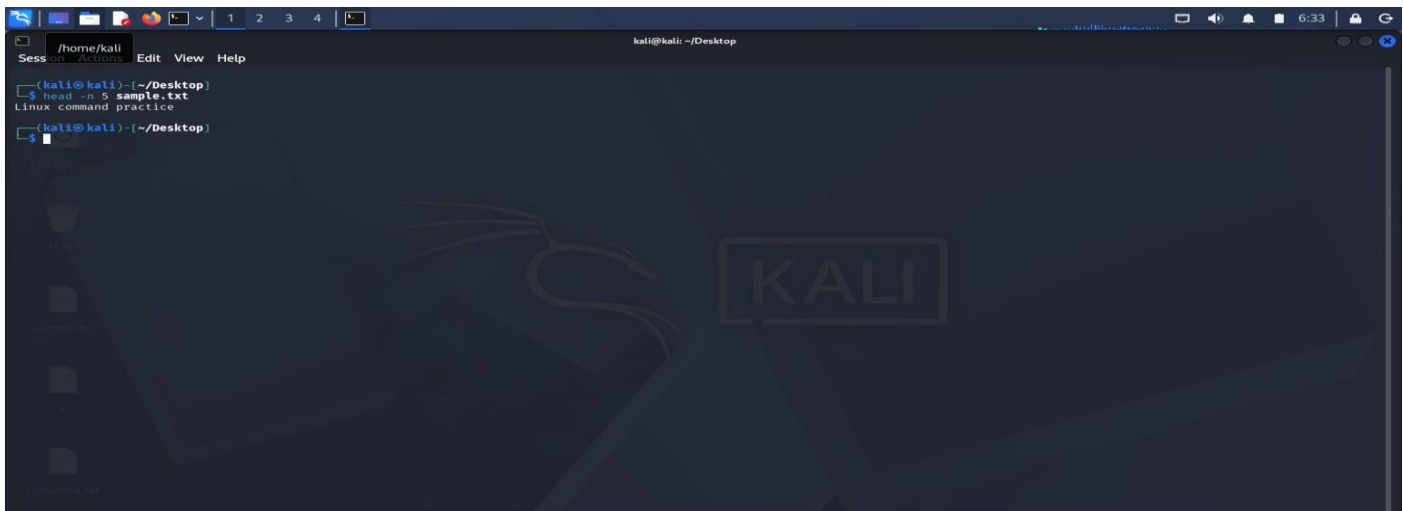
19. less :-

Brief: Allows you to view and navigate through a file interactively.



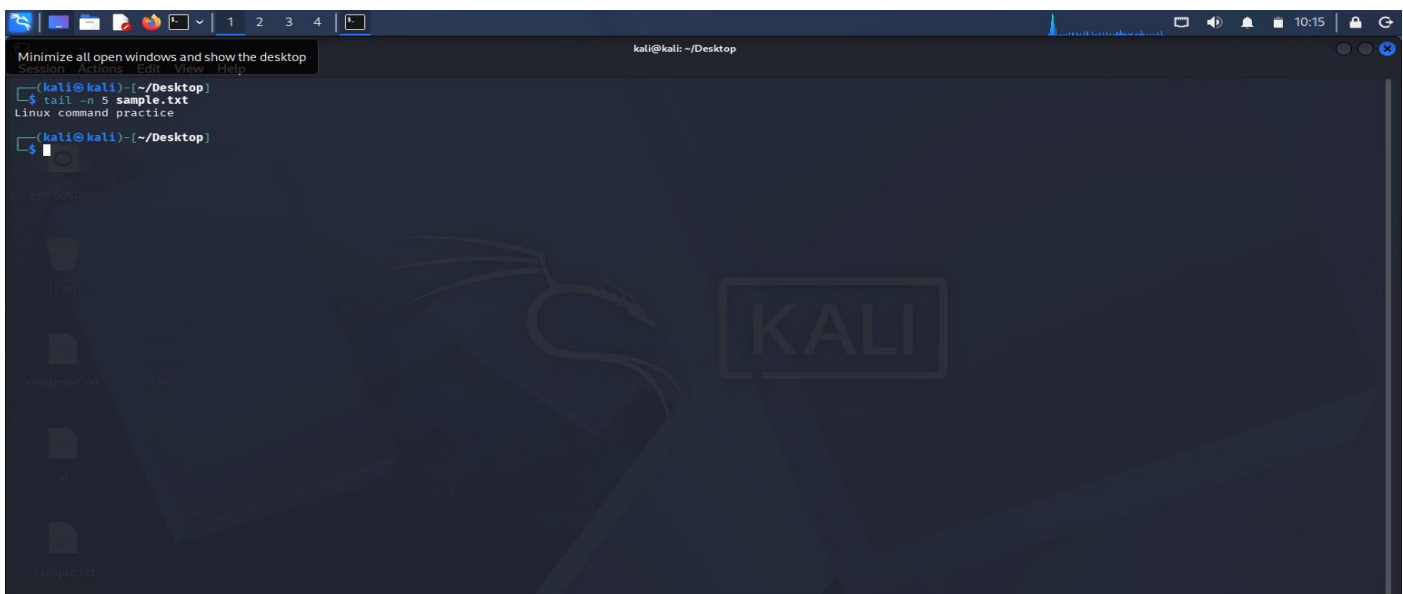
20. head :-

Brief: Displays the first lines of a file.



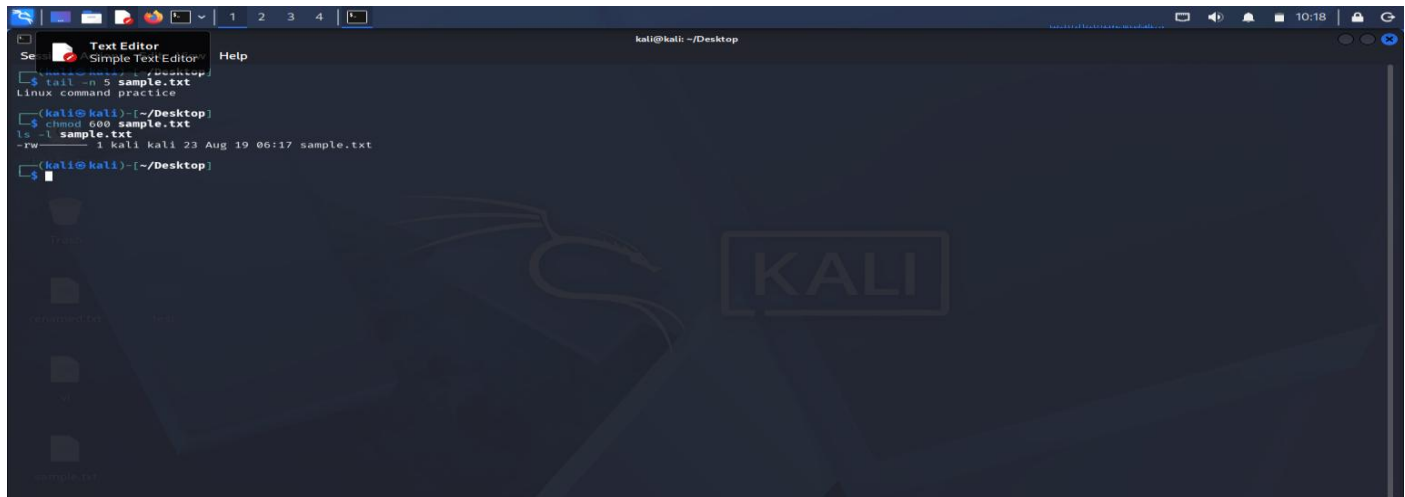
21. tail :-

Brief: Displays the last lines of a file.



22. chmod :-

Brief: Changes the permissions of a file.

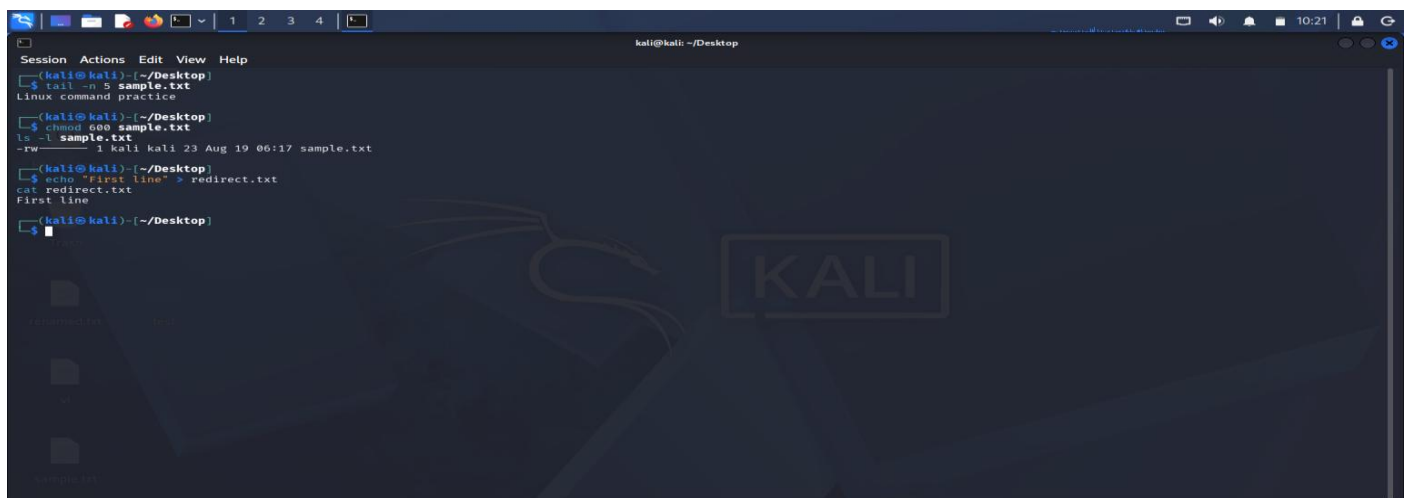
A terminal window titled 'kali@kali: ~/Desktop' showing the execution of the 'chmod' command. The user runs 'tail -n 5 sample.txt', then 'chmod 600 sample.txt', and finally 'ls -l sample.txt'. The output of the last command shows the file permissions as '-rw----- 1 kali kali 23 Aug 19 06:17 sample.txt'.

```
kali@kali: ~/Desktop
$ tail -n 5 sample.txt
Linux command practice

(kali@kali)~/Desktop
$ chmod 600 sample.txt
ls -l sample.txt
-rw----- 1 kali kali 23 Aug 19 06:17 sample.txt
(kali@kali)~/Desktop
$
```

23. > (OUTPUT REDIRECTION) :-

Brief: Sends command output to a file and overwrites existing contents.

A terminal window titled 'kali@kali: ~/Desktop' showing the use of output redirection. The user runs 'echo "First line" > redirect.txt' and then 'cat redirect.txt', which outputs 'First line'.

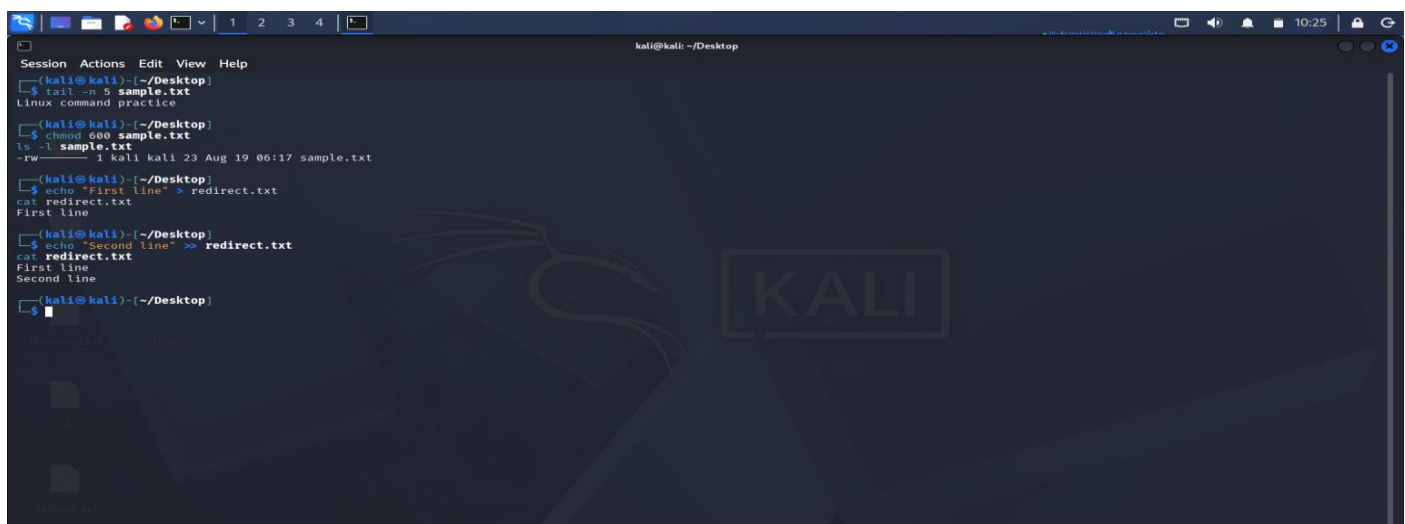
```
kali@kali: ~/Desktop
$ tail -n 5 sample.txt
Linux command practice

(kali@kali)~/Desktop
$ chmod 600 sample.txt
ls -l sample.txt
-rw----- 1 kali kali 23 Aug 19 06:17 sample.txt

(kali@kali)~/Desktop
$ echo "First line" > redirect.txt
cat redirect.txt
First line
(kali@kali)~/Desktop
$
```

24. >> (APPEND REDIRECTION) :-

Brief: Adds output to the end of a file without overwriting its existing contents.

A terminal window titled 'kali@kali: ~/Desktop' showing the use of append redirection. The user runs 'echo "First line" > redirect.txt', then 'cat redirect.txt' (outputting 'First line'), and finally 'echo "Second line" >> redirect.txt'. A subsequent 'cat redirect.txt' command outputs both 'First line' and 'Second line'.

```
kali@kali: ~/Desktop
$ tail -n 5 sample.txt
Linux command practice

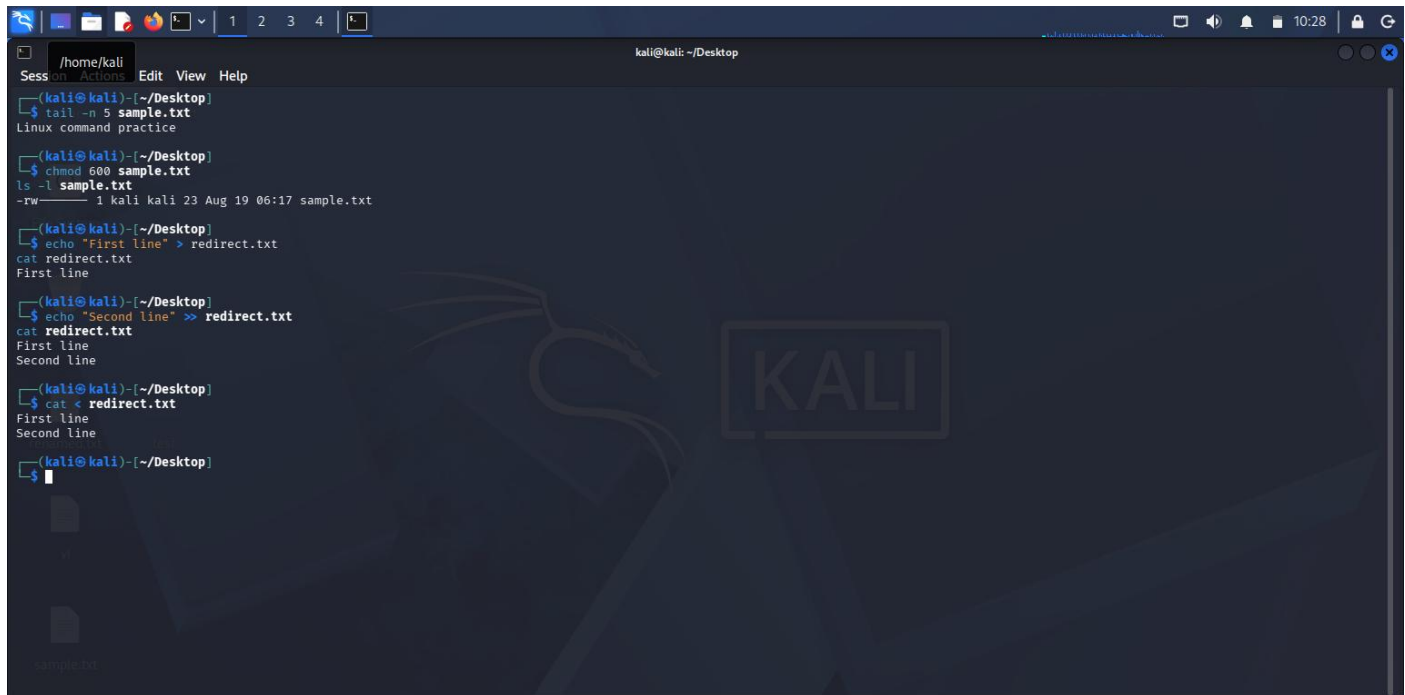
(kali@kali)~/Desktop
$ chmod 600 sample.txt
ls -l sample.txt
-rw----- 1 kali kali 23 Aug 19 06:17 sample.txt

(kali@kali)~/Desktop
$ echo "First line" > redirect.txt
cat redirect.txt
First line

(kali@kali)~/Desktop
$ echo "Second line" >> redirect.txt
cat redirect.txt
First line
Second line
(kali@kali)~/Desktop
$
```

25. < (INPUT REDIRECTION) :-

Brief: Takes input from a file and passes it to a command.



```
/home/kali
kali@kali: ~/Desktop
$ tail -n 5 sample.txt
Linux command practice

kali@kali:~/Desktop
$ chmod 600 sample.txt
ls -l sample.txt
-rw----- 1 kali kali 23 Aug 19 06:17 sample.txt

kali@kali:~/Desktop
$ echo "First line" > redirect.txt
cat redirect.txt
First line

kali@kali:~/Desktop
$ echo "Second line" >> redirect.txt
cat redirect.txt
First line
Second line

kali@kali:~/Desktop
$ cat < redirect.txt
First line
Second line

kali@kali:~/Desktop
$ less sample.txt
```
